

Malikeh Ehghaghi

LinkedIn: [Link](#) | Twitter: [Link](#) | Google Scholar: [Link](#)
Personal Website: [Link](#)

Toronto, ON, CA
Email: malikehehghaghi@gmail.com
Phone number: +1 (647) 9751375

Research Interests

- **LLM Safety & Security:** Safety evaluation and benchmarking, automated red-teaming, LLM safeguarding, and risk mitigation
- **Efficient & Scalable ML:** Decentralized training, modular AI, model merging, long-context learning, and test-time training
- **Trustworthy AI:** Factuality, interpretability, and fairness

EDUCATION

Ph.D. in Computer Science September 2025 - Present
Academic Advisor: Colin Raffel
University of Toronto, Department of Computer Science

MSc in Applied Computing September 2020 – December 2021
Academic Advisor: Frank Rudzicz
Industry Advisor: Jekaterina Novikova
University of Toronto, Department of Computer Science

BSc in Computer Engineering September 2015 – March 2020
Academic Advisor: Siamak Mohammadi
University of Tehran, Department of Electrical and Computer Engineering

EXPERIENCE

Vector Institute, Toronto, ON, CA February 2025 – Present
PhD Student (ML Research Scientist) - Supervisor: Colin Raffel

- **R3 Lab:**
 - Conducting research focused on decentralizing, democratizing, and de-risking large-scale AI.
 - Active Projects:
 - **FineSearch:** Reducing hallucinations in long-context QA through memory-efficient retrieval-augmented architectures trained on large-scale synthetic data.
 - **TokSuite:** Introducing a controlled framework for studying the impact of tokenization on language models by training identical models with different tokenizers. Introduces a real-world perturbation benchmark to analyze how tokenization strategies affect performance and robustness systematically.
 - **Risk-Under-Pressure:** Developing a compute-aware framework for evaluating LLM adversarial robustness, introducing FLOP-based risk metrics that quantify the computational effort required for successful jailbreak attacks beyond traditional attack success rates.
 - **PROTEUS:** Designing a Stackelberg game-theoretic defense framework that strategically randomizes LLM defense mechanisms to increase the computational cost of adaptive jailbreak and prompt injection attacks without requiring new defense algorithms.
 - **CAVEAT:** Designing a safety benchmark that reframes LLM safety evaluation from "Can an attack succeed?" to "Is the attack worth its computational cost?", combining attack utility and compute into a single decision-relevant metric.

Arcee AI, San Francisco, CA, US January 2024 – February 2025
Applied NLP Research Engineer

- **NLP Research Engineering Team:**
 - Engineering & Tooling:

- Contributed to Arcee's MergeKit (7K GitHub stars), the largest open-source model merging toolkit.
- Contributed to Arcee's DistilKit (~1K GitHub stars), a toolkit for distilling large teacher models into smaller, efficient ones.
- Built core components of Arcee's SaaS Platform, an end-to-end pipeline for efficient LLM training and inference.
- Contributed to developing Arcee Orchestra, an in-house platform for designing agentic workflows.
- Built multi-GPU training pipelines:

- On AWS Trainium for domain adaptation of LLMs.
 - On Together AI Cluster using Megatron-LM for training Mixture-of-Experts (MoE) models.
- o Project Leadership:
 - Contributed to building an automated workflow generation and large action planning for personal assistant bots in collaboration with SKTelecom
 - Built an agentic HR assistant bot in partnership with Riviera Partners
 - Built function-calling agent models for banking transactions in partnership with American Express (AmEx)
 - Implemented RAG-based Las Vegas tourism search system in partnership with MongoDB
 - Built KidsSafe LLMs with Angel Kids Inc., safe and personalized models for children aged 5–12
- o Multilingual & Social Impact Initiatives
 - Led the Arcee Globe initiative to build culturally and linguistically specific LLMs
 - Released Arcee Meraj and Meraj Mini, top-performing Arabic LLMs
- o Evangelism & Communication
 - Co-hosted the Small Language Models (SLMs) Show (with Julien Simon), covering SOTA in GenAI

Lavita, Palo Alto, CA, US

August 2023 – December 2023

Machine Learning Consultant

- **Machine Learning Research Team:**

- o Contributed to designing and implementing a framework for a general-purpose decentralized medical chatbot in partnership with Harvard Medical School and Dartmouth Center for Precision Health and Artificial Intelligence.

University of Toronto, Toronto, Canada

January 2023 – July 2023

Machine Learning Research Team Lead - Supervisor: Prof. En-Shiun Annie Lee

- **Lee Lab:**

- o Led a team of graduate and undergraduate students working on the medical interpretability of clinical notes
- o Benchmarked deep learning models to detect disease labels from clinical notes collected from electronic health records (EHRs)
- o Implemented model-agnostic and model-specific NLP interpretability techniques (e.g., SHAP, LIME, probing, adversarial techniques, etc.) to explore the most important input features in predicting the diagnosis labels
- o Benchmarked interpretability methods based on different interpretability evaluation techniques (e.g., human-centric evaluation, comprehensiveness, sensitivity, etc.)

Winterlight Labs Inc./Cambridge Cognition Inc., Toronto, Canada

April 2021 – July 2023

Machine Learning Engineer - Team Lead: Jekaterina Novikova

- **Machine Learning Research Team:**

- o Led client and research projects with a focus on monitoring and detection of cognitive impairment (e.g., Alzheimer's disease) and mental disorders (e.g., depression, Schizophrenia, anxiety, etc.) from speech and language data in multilingual settings in partnership with Johnson and Johnson and Genentech Roche Group.
- o Developed speaker verification tools for clinical trials to detect duplicate raters and participants with transferability to non-English languages (e.g., German, Danish, Spanish, and Arabic)
- o Analyzed the robustness of the disease detection models to different sources of noise, and developed tools to assess the interpretability and fairness of disease detection models for mental health disorders and cognitive impairment

KavoshComAsia Company, Tehran, Iran

February 2019 – February 2020

Machine Learning Research Assistant – Supervisor: Siamak Mohammadi

- **Research and Development Division:**

- o Designed and implemented on-device ECG heartbeat classification models for arrhythmia, applying deep convolutional neural networks and sequence-to-sequence models in the TensorFlow framework

York University, Toronto, Canada

Summer 2019

Machine Learning Research Intern – Supervisor: Prof. Aijun An

- **Data Mining Lab:**

- o Developed an unsupervised recommender system tool based on the shopping behavior and lifestyle of a group of people in collaboration with Manifold Data Mining Inc.

University of Tehran, Tehran, Iran

July 2018 – February 2019

Machine Learning Research Assistant – Supervisor: Prof. Mehdi Tale Masouleh

- **Human and Robot Interaction Laboratory:**

- o Implemented a motion planning and indoor localization system for Sanbot Robot, applying machine learning algorithms, including reinforcement learning, Particle Swarm Optimization (PSO), etc., in the TensorFlow framework

PEER-REVIEWED PUBLICATIONS

Altıntaş, Gül Sena*, **Ehghaghi, M.***, Lester, B., Liu, F., Zhao, W., Ciccone, M., Raffel, C., "TokSuite: Measuring the Impact of Tokenizer Choice on Language Model Behavior" - ICML 2026 (Spotlight)

Goddard, C., Siriwardhana, S., **Ehghaghi, M.**, Meyers, L., Karpukhin, K., Benedict, B., McQuade, M., Solawetz, J., "Arcee's MergeKit: A Toolkit for Merging Large Language Models" - EMNLP 2024 Industry Track - Accepted

Ehghaghi, M., Zhou, P., Rajabi, S., Cheng, W., Kuo, C., Lee, E., "Interpretable Disease Prediction from Clinical Text by Leveraging Pattern Disentanglement", IEEE BHI 2023 - Accepted

Ehghaghi, M., Stanojevic, M., Akram, A., and Novikova, J., "Factors Affecting the Performance of Automated Speaker Verification in Alzheimer's Disease Clinical Trials", 5th Clinical Natural Language Processing Workshop, ACL 2023 - Accepted

Ehghaghi, M., Rudzicz, F., and Novikova, J., "Data-driven Approach to Differentiating between Depression and Dementia from Noisy Speech and Language Data", 8th Workshop on Noisy User-generated Text, COLING 2022 - Accepted

Tasnim, M., **Ehghaghi, M.**, Diep, B., and Novikova, J., "Depac: a corpus for depression and anxiety detection from speech", 8th *Workshop on Computational Linguistics and Clinical Psychology*, NAACL 2022 - Accepted

Ehghaghi, M., Novikova, J., Sett, A., Hejrati, M., Robin, J., Teng, E., and Hashemifar S., "Benchmarking Prognostic Longitudinal Machine Learning Models of Alzheimer's Disease Using Speech Features", 1st Workshop on Applications of Medical AI, MICCAI 2022 - Accepted

PREPRINTS

Ehghaghi, M.*, Ecsedi, B.*, Chechik, M., Raffel, C., "Risk Under Pressure: Compute-Aware Evaluation of Adversarial Robustness in Language Models" - Available on arxiv.org (Under Review for NeurIPS 2026)

Akram, A., **Ehghaghi, M.**, Stanojevic, M., and Novikova, J., "Zero-Shot Multilingual Speaker Verification in Clinical Trials" - Available on arxiv.org

Gauthier-Caron, T., Siriwardhana, S., Stein, S., **Ehghaghi, M.**, Goddard, C., McQuade, M., Solawetz, J., Labonne, M., "Merging in a Bottle: Differentiable Adaptive Merging (DAM) and the Path from Averaging to Automation" - Available on arxiv.org

WORKSHOPS

- The First Workshop on Machine Learning for Cognitive and Mental Health (ML4CMH), AAAI 2024 - Program Co-Chair

INVITED TALKS

- "TokSuite: Measuring the Impact of Tokenizer Choice on Language Model Behavior" paper at ICML 2026 - Presenter
- "Women in Industry Research", Build With AI: Women in Tech (WIT) conference 2026 by Google Developer Groups (University of Toronto) and ACM-W: ACM's Women in Computing - Panelist
- "Model Merging: Theory, Practice and Applications", In-person Tutorial, NeurIPS 2025 - Speaker
- "Scaling Down, Powering Up: Can Efficient Training Beat Scaling Laws", In-person Workshop, Toronto Machine Learning Summit (TMLS) 2025 - Speaker
- Panel Discussions with Female Leaders in the GTA in Tech, Girls in Tech Conference (GITCon) 2025 - Panelist
- "From Bias to Brilliance: How Women Are Rewriting the Rules of AI", Women in Tech (WIT) Conference 2025 by Google Developer Groups (GDG) and ACM-W: ACM's Women in Computing - Speaker
- Women in AI Research (WiAIR) Podcast, 2025 - Co-host
- "Factors Affecting the Performance of Automated Speaker Verification in Alzheimer's Disease Clinical Trials" paper at 5th Clinical Natural Language Processing Workshop, ACL 2023 - Presenter
- "Data-driven Approach to Differentiating between Depression and Dementia from Noisy Speech and Language Data" paper at 8th Workshop on Noisy User-generated Text, COLING 2022 - Presenter
- Wo(Men) In Tech Program, Google Developer Student Clubs, Fall 2021 - Industry speaker

TEACHING EXPERIENCE

FastLane Machine Learning Associates (MLA) Cohort, Vector Institute, Toronto, Canada Technical facilitator	Summer 2026
Data Readiness, Model Development & Deployment (DaRMod), Vector Institute, Toronto, Canada Technical facilitator	Summer 2026
Agentic AI Bootcamp 1.0 and 2.0, Vector Institute, Toronto, Canada Technical facilitator	Summer 2025 - Winter 2026
Agentic AI Evaluation Bootcamp 1.0, Vector Institute, Toronto, Canada Technical facilitator	Winter 2026

University of Toronto in Partnership with Vector Institute, Toronto, Canada Instructional Assistant	March 2023 – April 2023
TUSK Upskilling Series: Machine Learning Software Foundations - Winter 2023, Spring 2023	

- University of Toronto, Toronto, Canada**
Graduate Teaching Assistant
- CSC413/2516: Neural Networks and Deep Learning (Prep TA) - Summer 2026
 - CSC413/2516: Neural Networks and Deep Learning - Fall 2025
 - CSC457H1: Principles of Computer Networks - Fall 2020
 - CSC343: Introduction to Databases - Spring 2021
 - CSC263: Data Structures and Analysis - Fall 2021

TECHNICAL SKILLS

Languages: Python (Expert) • Java • R • C • C++ • SQL (Familiar)

NLP/AI Specializations: LLMs • Transformers • Model Merging • Pruning • Distillation • Reward Modeling • Reinforcement Learning • Sequence Modeling • RNN • LSTM • GAN • Encoder-Decoder Models • LLM Safety and Security • Interpretability and Fairness • LLM Evaluation and Benchmarking • Language Modeling • Mixture of Agents • Mixture of Experts (MoE) • Retrieval-Augmented Generation (RAG) • Continual Pretraining • Multilingual Learning • Multimodal Learning • Automated Red-teaming • LLM Safeguards • LLM Pretraining and Post-training • Modular AI • Decentralized Training

Packages: PyTorch • Huggingface Transformers • Scikit-Learn • LangChain • LangGraph • Megatron-LM • vLLM • Meta Lingua • Alignment Handbook • Axolotl • NumPy • Pandas • SpaCy • NLTK • Matplotlib/Seaborn • BeautifulSoup • DataTrove • Open AI Agent SDK • Dify • Axolotl • LM Eval Harness • Stanza • Stanford NLP • Google Lens • Tesseract • LlamaParse • Unstructured

Cloud/DevOps: AWS (SageMaker, Trainium, Glue, EC2, ECR) • PySpark • ClearML • CircleCI • Runpod • Docker • Git • Weights and Bias • Claude Code • Langfuse • E2B Sanbox • MCP

HONORS and AWARDS

- Faculty of Arts & Science Top (FAST) Doctoral Award, University of Toronto, 2025
- Vector PhD Research Grant, Vector Institute, 2025
- MITACS Accelerate Scholarship, University of Toronto, 2021
- Vector Scholarship in Artificial Intelligence, University of Waterloo, 2020 [DECLINED]
- Ranked 1st (Highest GPA) among all students of BSc in computer engineering entered in 2015, University of Tehran, Iran

Volunteer Services

- Community Support:
- Vice Chair, ACM-W Professional Chapter (since September 2025)
 - Free Mentorship Program for Juniors in Tech (since 2023)
 - MScAC Mentorship Program, Mentor (2023–2024 cohort)
- Conference & Workshop Reviewing:
- Paper Reviewer, NeurIPS 2026 (Evaluations and Datasets Track)
 - Paper Reviewer, ICLR 2026 Workshop on Principled Design for Trustworthy AI - Interpretability, Robustness, and Safety across Modalities
 - Paper Reviewer, NeurIPS 2025 (Position Paper Track)

- Paper Reviewer, ICLR 2025 Workshop on Modularity for Collaborative, Decentralized, and Continual Deep Learning
- Paper Reviewer, AAAI 2024 Workshop on Responsible Language Models

Languages

- Farsi - Native proficiency
- English - Professional working proficiency
- Arabic - Limited working proficiency
- French - Limited working proficiency